

model-access-architecture

Model-Access Architecture — Diagrams (Key-Recognition · Unified Catalog · CLI-Bridge)

Field	Value
Revision	1
Created	2026-06-10
Last modified	2026-06-10
Status	DESIGN — describes target architecture; nodes labelled IMPLEMENTED vs PLANNED
Maintainer	DOCUMENTATION subagent (READ-ONLY on code)
Authority	Cascades from constitution/ + root CLAUDE.md/CONSTITUTION.md; CONST-036 (verifier single source of truth), CONST-040 (capabilities from VerificationResult), §11.4 anti-bluff
Sources	SP1 plan (docs/superpowers/specs/plans/2026-06-10-SP1-model-access-plan.md), SP2 plan (.../2026-06-10-SP2-helixagent-exposure-plan.md), SP4 plan (.../2026-06-10-SP4-cli-bridge-plan.md); analyses B/C/D (docs/superpowers/specs/analysis/2026-06-10-{B,C,D}-*.md); master roadmap (docs/superpowers/specs/2026-06-10-llms-access-master-roadmap.md)

§11.4 / §11.4.123 anti-bluff honesty note (read first). This document describes the **target architecture** of the model-access + provider-exposure + CLI-bridge system. It mixes **IMPLEMENTED** components (real code, cited file:line, confirmed in the analysis passes) with **PLANNED** components (specified in SP1/SP2/SP4 but **not yet built** — no captured runtime evidence exists). Every diagram node carries an explicit [IMPLEMENTED ...] or [PLANNED ...] tag. A node tagged PLANNED is a design intent, **not** a claim of working software. Diagrams are documentation, not proof of behaviour; the runtime-evidence bar (§11.4.5 / §11.4.69 / §11.4.107) is met only by the test/Challenge evidence each SP plan requires at execution time.

Legend used in every diagram:

- **IMPLEMENTED** — real code exists today; cited to a real file:line read during the 2026-06-10 analysis passes.
- **PLANNED** — specified in an SP plan / task (e.g. SP1·T1.2.1); **not built**; no runtime evidence.
- **DEAD/UNWIRED** — code exists but has zero production call-sites (a §11.4.124 investigate-then-wire target, **not** removed).

Paths: inner Go app code is under `helix_code/` (module `dev.helix.code`); the HelixAgent exposure surface is under `submodules/helix_agent/` (module `dev.helix.agent`).

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Diagram A — Key-recognition → verifier → working-model funnel (SP1)

What it shows. How a single supplied/recognized API key turns into the set of WORKING models a user sees. Source: SP1 plan §3 “Target design (the working-model funnel)”; analysis-B §4.1.

Working-model predicate (SP1 §3, applied in the PLANNED GetWorkingModels):
`present[provider] ∧ Verified==true ∧`
`VerificationStatus=="verified" ∧ OverallScore >=`
`GetMinAcceptableScore() ∧ Providers[provider].Enabled != false.`

flowchart TD

```
subgraph STARTUP["Process startup (cmd/server/main.go AND cmd/cli/main.go)"]
    A1["secrets.LoadAPIKeys()<br/>reads $HOME/api_keys.sh then walked-up"]
    A2["config.Load() (viper AutomaticEnv + SetEnvPrefix HELIX)<br/>8"]
    A3["keyrec.PresentProviders() map[string]bool<br/>multi-alias {provider}"]
end
```

```
subgraph KEYSRC["Key sources (any one ⇒ provider present)"]
    K1["$HOME/api_keys.sh (export VAR=val)<br/>[IMPLEMENTED reader — loader.go:84-129]"]
    K2["walked-up .env<br/>[IMPLEMENTED reader — loader.go:84-129]"]
    K3["shell-exported env (OPENAI_API_KEY, ANTHROPIC_API_KEY, ...)"]
end
```

```
subgraph VERIFIER["LLMsVerifier integration (CONST-036 single source of truth)"]
    V1["verifier.Adapter.GetVerifiedModels(ctx)<br/>cache -> live HTTP"]
    V2["GetMinAcceptableScore() default 6.0 (HELIX_VERIFIER_MIN_SCORE)"]
    V3["GetWorkingModels(ctx, present)<br/>applies key-presence + VerificationStatus"]
end
```

```

    V4["filterByProviderConfig — drops only Enabled==false<br/><b>[IMP
end

subgraph EXPOSE["Exposure call-sites"]
    E1["CLI handleListModels<br/>P1 verifier GetVerifiedModels -> prin
    E2["server handlers /api/models<br/><b>[IMPLEMENTED -> GetVerified
    E3["model_manager registry<br/>SetVerifierAdapter exists; registry
end

subgraph FACTORY["Provider construction"]
    F1["factory.NewProvider(config) — builds ONE provider by config.Ty
    F2["BuildPresentProviders(configs, present) — build EVERY key-pres
    F3["per-provider GetModels() — OpenAI/Anthropic/DeepSeek/Mistral H
end

DISABLED["Verifier disabled / cold (HELIX_VERIFIER_ENABLED=false defau

K1 --> A1
K2 --> A1
A1 --> A2
K3 --> A2
A2 --> A3
A3 --> V3
V1 --> V3
V2 --> V3
V3 --> V4
V4 --> E1
V4 --> E2
V4 --> E3
A3 --> F2
F1 --> F2
F2 --> F3
A2 -. "verifier off" .-> DISABLED
DISABLED -. "honest list" .-> E1

```

Defect anchors closed by SP1 (analysis-B top-5 gaps): D-2 (CLI shows failed/pending), D-3 (LoadAPIKeys dead), D-4 (min-score never applied), D-5 (hardcoded lists), plus the no-key-presence-gate gap and the single-provider-factory gap.

Diagram B — Unified catalog under one root (SP2 + SP4)

What it shows. The PLANNED unified catalog that joins, **under one root**, the AI-debate ensemble + HelixLLM + every discovered provider + every provider's verified model + (SP4) every CLI-agent bridge provider and its models — each a uniformly-named, individually-addressable selection target. Source: SP2 plan §P2.1/§P2.2; analysis-C §3; master roadmap §257; SP4 plan §P4.1/§P4.3.

Naming grammar (analysis-C §3.2, master roadmap line 257): ensemble · ensemble / <preset> · helixllm · helixllm / <model> · <provider> · <provider> / <model_id> (preserving already-namespaced ids like openrouter / x-ai / grok-4) · (SP4) cli / <agent> · cli / <agent> / <model>.

flowchart LR

```
subgraph SOURCES["Join sources (all IMPLEMENTED individually)"]
    S1["ProviderRegistry.ListProviders() + GetCapabilities().Supported"]
    S2["verifier ModelDiscoveryService<br/>DiscoveredModel.Verified /"]
    S3["Ensemble service GetEnsembleService().RunEnsemble<br/><b>[IMPL"]
    S4["HelixLLM gated provider 'helixllm' (USE_HELIX_LLM=true)<br/><b>["]
    S5["CLI-agent bridge providers (claude/qwen/encode/...)<br/><b>["]
end

subgraph CATALOG["Unified catalog (the missing join layer) — ONE ROOT"]
    C1["catalog.BuildCatalog(reg, verifier, ensembleSvc[, cliRegistry]"]
    C2["CatalogHandler.List -> GET /v1/catalog<br/><b>[PLANNED — SP2·T"]
    C3["Selector resolution: any catalog Name -> target<br/>unknown ->"]
end

subgraph NAMES["Uniformly-named selection targets (one namespace)"]
    N0["ensemble<br/>ensemble/confidence_weighted"]
    N1["helixllm<br/>helixllm/helixllm-default"]
    N2["anthropic<br/>anthropic/claude-3-sonnet-20240229"]
    N3["openrouter/x-ai/grok-4 (preserved namespacing)"]
    N4["cli/claude<br/>cli/claude/&lt;model&gt; (SP4 PLANNED)"]
end

BLUFF["GET /v1/completion/models — HARDCODED 3-model list (D-1)<br/>de"]

S1 --> C1
S2 --> C1
S3 --> C1
S4 --> C1
S5 --> C1
C1 --> C2
C2 --> C3
C1 --> N0
C1 --> N1
C1 --> N2
C1 --> N3
C1 --> N4
BLUFF -. "remediated, joins same source" .-> C1
```

Anti-bluff invariant (SP2 §7). Catalog “model” entries are emitted **only** for `DiscoveredModel.Verified == true` — never the static `SupportedModels` slice, never a hardcoded literal. HelixLLM is **promoted, never removed** (§11.4.122); all existing `/v1/*` routes stay intact (additive design).

Diagram C — CLI-bridge selection & proxy sequence (SP4)

What it shows. How a `cli/<agent>` selection resolves a binary (PRIMARY system-installed → FALLBACK submodule-built), proxies a prompt via real `exec`, parses the result, and surfaces power-features. Source: SP4 plan §P4.2/§P4.3/§P4.4; analysis-D §5.

Tier-1 agents system-installed today (analysis-D §2, captured command -v): claude 2.1.170, qwen 0.5.0, opencode 1.16.2, gemini 0.1.9, crush v0.22.2, codex, goose 1.8.0, copilot 1.0.46.
Fallback-required (absent today): aider, plandex, forge.

sequenceDiagram

```
autonumber
participant U as User / catalog selector (cli/&lt;agent&gt;)
participant P as CLIAgentProvider (llm.Provider)<br/>[PLANNED — SP4·T4
participant R as resolveBinary(spec)<br/>[PLANNED — SP4·T4.2.1]
participant SYS as System PATH (exec.LookPath)<br/>[IMPLEMENTED primit
participant FB as Submodule-built binary<br/>cli_agents/&lt;agent&gt;/
participant EX as exec.CommandContext(binary, non-interactive args)<br/>
participant V as LLMsVerifier VerificationResult<br/>[IMPLEMENTED — CO
```

U->>P: Generate(ctx, req) / GetModels() / GetCapabilities()

P->>R: resolve binary for AgentSpec

R->>SYS: exec.LookPath(spec.Binary) — PRIMARY

alt PRIMARY found (tier-1: claude/qwen/opencode/...)

SYS-->>R: /path/to/binary

else PRIMARY absent (aider/plandex/forge)

R->>FB: resolve submodule-built artifact (FALLBACK)

FB-->>R: built bin path

else both absent

R-->>P: IsAvailable=false + SKIP-with-reason (binary_unavailable)<

end

R-->>P: chosen path (recorded in GetHealth — anti-bluff provenance)

Note over P,EX: NEVER the qwencode APIKey!="" bluff (qwencode.go:137)<

P->>EX: prompt via stdin / --prompt; non-interactive flag (§11.4.99 pe

EX-->>P: real stdout + real exit code

P->>P: parse stdout -> LLMResponse (GenerateStream = line-buffered)

par Power-features + dynamic models

P->>EX: model-list subcommand -> parse -> []ModelInfo (CONST-036,

P->>V: GetCapabilities() from VerificationResult (Vision/RAG/Memor

end

Note over P: Config re-export -> INSTALL (backup-first §9.2) -> LIVE p

P-->>U: real result (USER-VISIBLE proof, §11.4.108 runtime signature:

Wiring prerequisite (SP4 §2.3, confirmed gap D-7). helix_code/go.mod (replace block lines 194-212) has **no** replace dev.helix.agent — so HelixCode cannot import the cli substrate today. SP4·T4.0.5 PLANS to add it (or a narrow shim, decided on go mod graph evidence). The exec primitive itself (base.go:120-138) is IMPLEMENTED and real; the per-agent dispatch tier (instance_manager.go:906+ executeClaudeCode/...) and qwencode.go:101/115/137 are **STUBS** (a §11.4 bluff SP4·P4.0 RED-reproduces then fixes).

Node provenance ledger (IMPLEMENTED vs PLANNED)

A single-table audit so a reader can verify each claim against real code or a real SP task.

Component	State	Anchor (real file:line) or SP
secrets.LoadAPIKeys reader	IMPLEMENTED but DEAD/ UNWIRED	helix_code/internal/sec loader.go:30,33-37,84-1
Wire LoadAPIKeys at startup	PLANNED	SP1·T1.1.1 (cmd/server/main main.go)
Viper AutomaticEnv + 8 BindEnv	IMPLEMENTED	helix_code/internal/con config.go:307-345
Multi-alias key-recognition table	PLANNED	SP1·T1.1.2 (internal/llm/ keyrecognition.go)
helix_agent SupportedProviders.EnvVars (reuse template)	IMPLEMENTED	submodules/helix_agent/ verifier/provider_types
verifier.Adapter.GetVerifiedModels (real HTTP)	IMPLEMENTED	helix_code/internal/ver adapter.go:183-224; clie api/models
GetMinAcceptableScore (loaded, never invoked)	IMPLEMENTED- unused (D-4)	helix_code/internal/ver adapter.go:175-180
GetWorkingModels filter	PLANNED	SP1·T1.2.1 / T1.3.1 (adapter.g
filterByProviderConfig (Enabled flag only)	IMPLEMENTED	helix_code/internal/ver adapter.go:277-289
CLI handleListModels / printVerifiedModels (shows failed/ pending, D-2)	IMPLEMENTED- bluffy	helix_code/cmd/cli/ main.go:1355-1414
Switch CLI/server/registry → GetWorkingModels	PLANNED	SP1·T1.3.2
Verifier-disabled honest behaviour	PLANNED	SP1·T1.3.3 / DECISION-2 (today main.go:1387)
factory.NewProvider (one by config.Type)	IMPLEMENTED	helix_code/internal/llm factory.go:9-101
BuildPresentProviders enumerator	PLANNED	SP1·T1.2.2
Hardcoded provider model lists (D-5/ CONST-036)	IMPLEMENTED (violation)	openai_provider.go:201, anthropic_provider.go:2 deepseek_provider.go, mistral_provider.go
OpenRouter fetchCatalog / Ollama / api / tags (dynamic templates)	IMPLEMENTED	openrouter_provider.go: ollama_provider.go:225
ProviderRegistry.ListProviders + GET /v1/providers	IMPLEMENTED	submodules/helix_agent/ services/provider_regis internal/router/router.
Ensemble route POST /v1/ensemble/ completions	IMPLEMENTED	submodules/helix_agent/ router/router.go:676-76
HelixLLM gated provider helixllm (buried)	IMPLEMENTED	

Component	State	Anchor (real file:line) or SP
		submodules/helix_agent/ services/ provider_registry.go:75
DiscoveredModel.Verified / OverallScore (working flag)	IMPLEMENTED	submodules/helix_agent/ verifier/discovery.go:5
GET /v1/completion/models hardcoded 3-model list (D-1)	IMPLEMENTED bluff	submodules/helix_agent/ handlers/completion.go:
D-1 verifier-sourced rewrite	PLANNED	SP2·T2.0.2
catalog.BuildCatalog join	PLANNED	SP2·T2.1.2 (internal/catalog catalog.go)
GET /v1/catalog handler + route	PLANNED	SP2·T2.1.3 (OP-1 root path)
Selector resolution (Name → target)	PLANNED	SP2·T2.1.4
HelixLLM first-class root promotion	PLANNED	SP2·T2.2.4
llm.Provider contract (CLI bridge target)	IMPLEMENTED	helix_code/internal/llm missing_types.go:356-37
CLI-exec primitive (LookPath + CommandContext)	IMPLEMENTED	submodules/helix_agent/ clis/agents/base/base.g
qwencode stub (templated strings, APIKey gate)	IMPLEMENTED bluff (D-6)	submodules/helix_agent/ clis/agents/qwencode/ qwencode.go:101,115,137
instance_manager dispatch stubs	IMPLEMENTED bluff (wider than D-6)	submodules/helix_agent/ clis/instance_manager.g
replace dev.helix.agent wiring (D-7)	PLANNED/ ABSENT	SP4·T4.0.5 (helix_code/go.m has no such replace)
CLIAgentProvider + AgentSpec registry	PLANNED	SP4·T4.1.2 / T4.1.3
resolveBinary PRIMARY→FALLBACK	PLANNED	SP4·T4.2.1
Dynamic GetModels per agent (CONST-036)	PLANNED	SP4·T4.3.1
Capabilities from VerificationResult (CONST-040)	PLANNED	SP4·T4.3.2
Config re-export + static Validate()	IMPLEMENTED	submodules/helix_agent/ unified_cli_generator.g cmd/helixagent/main.go:
Filesystem INSTALL + LIVE post-install validate	PLANNED/ ABSENT	SP4·T4.4.2 / T4.4.3

Cross-references

- **SQL data model** for these flows: [model-access-schema.sql](#).
- **Index / metadata header:** [README.md](#).

- **SP plans (authoritative task source):** docs/superpowers/specs/plans/2026-06-10-SP1-model-access-plan.md, ...-SP2-helixagent-exposure-plan.md, ...-SP4-cli-bridge-plan.md.
- **Analysis (evidence base):** docs/superpowers/specs/analysis/2026-06-10-B-model-access.md, ...-C-helixagent-exposure.md, ...-D-cli-bridge.md.
- **Master roadmap:** docs/superpowers/specs/2026-06-10-llms-access-master-roadmap.md.

Per CONST-066 / §11.4.65, this .md should ship synchronized .html/.pdf siblings at release-prep. They are **not** generated by this READ-ONLY documentation pass.